

ASG CAD MASTER STANDARD - AutoCAD Environment for Aluminium, Glass & Facade Engineering

Standard R1 | 10.10.2026 | Supersedes ASG-STD-CAD-001 R0 and the earlier cad-standard / asg-master-template packages. Conventions are informed by ISO 128 (line widths), ISO 3098 (lettering heights), ISO 216 (paper sizes), ISO 5457 (borders), ISO 5455 (scales) and ISO 7200 (title block data). No formal compliance or certification is claimed - the ISO texts were not available for verification.

1. Design philosophy

1. Model geometry is drawn full size, 1:1, in millimetres.
2. Geometry lives in model space; annotation is annotative in model space or true-size in paper space; sheets live in layouts.
3. Layers are meaningful, coded and documented; object properties are ByLayer.
4. Blocks have predictable names, datum base points and mm insertion units.
5. Dimensions always report true geometry - dimension text is never overridden.
6. Annotation is sized by its plotted paper height (ISO 3098 series), never guessed in model space.
7. Layouts use genuine ISO 216 paper sizes and locked, controlled viewports.
8. Plots are monochrome with a fixed ISO 128 lineweight hierarchy - predictable on every printer and PDF.
9. Proprietary manufacturer data is never invented.
10. Easy to learn, hard to misuse: few styles, clear names, one setup command, one QA command.

2. Package structure

Folder	Content
01_Master_Template	ASG_Master_Template.dxf (master) + ASG_Setup.lsp (setup & QA). DWG / DWT are saved here from AutoCAD (README step 5).
02_Drawing_Templates	One single-layout template per sheet size (DXF)
03_Page_Setups	ASG_Page_Setups.csv - page setup definitions (created as named page setups by ASG-SETUP)
04_Plotting_Standards	ASG_Monochrome.ctb
05_CAD_Standards_Documentation	ASG_CAD_Standards.pdf (this document), ASG_CAD_Quick_SOP.pdf (one-page drafting SOP for every draftsman) + layout previews
06_Layer_Register	ASG_Layer_Register.xlsx / .csv
07_QA_Validation	Validation report PDF / TXT, temporary QA drawing, QA renders
08_Automation_Source	asg_standard.py (all definitions), build_all.py, build_sop.py, qa_validate.py

3. Units and drawing-resident variables written into the template

Variable	Value	Purpose
INSUNITS	4	Insertion units = millimetres (no unit conversion on block insert)
MEASUREMENT	1	Metric - acadiso.lin / acadiso.pat
LUNITS	2	Linear units = Decimal
LUPREC	0	Linear display precision 0 dp
AUNITS	0	Angular units = Decimal Degrees
AUPREC	2	Angle display precision 0.00
ANGBASE	0.0	0 deg = East
ANGDIR	0	Counter-clockwise positive
LIMMIN	(0.0, 0.0)	Nominal limits origin
LIMMAX	(200000.0, 150000.0)	Nominal 200 x 150 m - never restrictive
LIMCHECK	0	Limits not enforced (large facade elevations)
LTSCALE	1.0	Global linetype scale 1 - dash lengths in paper mm
PSLTSCALE	1	Viewport scale controls paper-space linetype scaling
CELTSSCALE	1.0	New-object linetype scale 1
LWDISPLAY	1	Lineweights displayed on screen
CELWEIGHT	-1	New objects: lineweight ByLayer
CECOLOR	256	New objects: colour ByLayer
DIMASSOC	2	Associative dimensions follow geometry
PLINEGEN	1	Continuous linetype pattern along polylines
FILLMODE	1	Fills displayed
MIRRTEXT	0	Text not mirrored
TEXTSIZE	2.5	Default text height (paper mm for annotative styles)
PSTYLEMODE	1	Colour-dependent plot styles (CTB)
WORLDVIEW	1	UCS follows WCS in views
UCSORG	(0.0, 0.0, 0.0)	UCS origin = WCS origin
UCSXDIR	(1.0, 0.0, 0.0)	UCS = World
UCSYDIR	(0.0, 1.0, 0.0)	UCS = World
VISRETAIN	1	Xref layer overrides retained
PROXYGRAPHICS	1	Proxy images saved (interoperability)
CLAYER	ASG-OUTLINE-PRIMARY	Default current layer (plottable geometry)
TEXTSTYLE	ASG-TEXT-ANNO	Default text style
DIMSTYLE	ASG-DIM-ANNO	Default dimension style (header dim vars synchronised - no overrides)
CMLSTYLE	ASG-MLINE-2	Default multiline style

Drawing-resident variables set inside AutoCAD by ASG-SETUP (saved with the DWG / DWT):

Variable	Value	Purpose
CANNOSCALE	1:1	Current annotation scale in model space
MSLTSCALE	1	Model-space linetypes follow the annotation scale
ANNOALLVISIBLE	0	Show only annotation that supports the current scale (no visual duplicates)
CMLEADERSTYLE	ASG-ML-ANNO	Default multileader style
CTABLESTYLE	ASG-TABLE-STANDARD	Default table style
HPLAYER	ASG-HATCH	New hatches go to ASG-HATCH automatically

Variable	Value	Purpose
DIMLAYER	ASG-DIMENSION	New dimensions go to ASG-DIMENSION automatically
UCSFOLLOW	0	View does not rotate with UCS changes

Precision: ordinary dimensions 0 dp; fabrication details ASG-DIM-DETAIL 0.0; angles 0.00. INSUNITS = mm on the drawing and every block - no unit conversion on insertion.

4. Machine / user-profile settings (NOT stored in any DWG / DWT)

These live in each PC's AutoCAD profile (registry). A template cannot enforce them. Recommended values, applied only by the opt-in command ASG-PROFILE.

Variable	Recommended	Purpose
OSMODE	2223	Endpoint, Midpoint, Centre, Node, Intersection, Perpendicular, Extension
AUTOSNAP	63	Marker, magnet, tooltip, aperture, polar + object snap tracking
POLARANG	45.0	Polar tracking increment 45 deg
ORTHOMODE	0	Ortho off - polar tracking preferred
DYNMODE	3	Dynamic input: pointer + dimension input
DYNPROMPT	1	Prompts near cursor
SELECTIONCYCLING	2	Cycle overlapping objects with list
GRIPS	1	Grips on
GRIPSIZE	5	Grip size
PICKBOX	4	Selection target size
APERTURE	8	Object snap target size
PICKFIRST	1	Noun-verb selection
PICKADD	2	Shift removes from selection
ANNOAUTOSCALE	0	Do NOT auto-add scales to annotative objects
SAVETIME	10	Autosave every 10 min
ISAVEBAK	1	Keep .bak on save
LAYOUTREGENCTL	2	Fast layout switching
REFPATHTYPE	1	Relative xref paths

5. Layer dictionary

Naming: ASG-<FUNCTION>[-<QUALIFIER>] in full words. Codes: G geometry, T technical, A annotation, S sheet. All objects ByLayer. Layer 0 for block definitions only. Defpoints is an AutoCAD system layer (never plots).

Code	Layer	ACI	Linetype	LW	Plot	Recommended use
G01	ASG-OUTLINE-PRIMARY	7	Continuous	0.35	Yes	Overall frame / panel / opening outlines in elevation and plan
G02	ASG-OUTLINE-SECONDARY	8	Continuous	0.18	Yes	Edges beyond, minor returns, internal lines
G03	ASG-ALUMINIUM-PROFILE	7	Continuous	0.25	Yes	Frames, mullions, transoms, sashes, beads (elevation and section)
G04	ASG-GLASS	4	Continuous	0.18	Yes	Glass / mirror edges, DGU and laminate ply lines
G05	ASG-STAINLESS-STEEL	7	Continuous	0.25	Yes	SS fittings, spigots, channels, handrails, patch plates
G06	ASG-MILD-STEEL	7	Continuous	0.25	Yes	MS brackets, sub-frames, support steel, base plates
G07	ASG-HARDWARE	8	Continuous	0.18	Yes	Hinges, handles, locks, rollers, closers, patch fittings
G08	ASG-FIXING	8	Continuous	0.18	Yes	Anchors, screws, bolts, rivets, fixing brackets
G09	ASG-GASKET	8	Continuous	0.13	Yes	EPDM gaskets, brushes, setting / distance blocks
G10	ASG-SEALANT	8	Continuous	0.13	Yes	Structural / weather silicone, backer rod, tapes
G11	ASG-INSULATION	252	Continuous	0.13	Yes	Insulation and thermal breaks (plots grey)
G12	ASG-CLADDING	7	Continuous	0.25	Yes	ACP / solid panels, flashings, copings
T01	ASG-HIDDEN	8	HIDDEN2	0.18	Yes	Concealed edges, embedded parts
T02	ASG-CENTERLINE	8	CENTER2	0.13	Yes	Axes of symmetry, hole centres, module lines
T03	ASG-CUTTING-PLANE	7	PHANTOM2	0.35	Yes	Section cut lines (section markers go on ASG-SYMBOL)
T04	ASG-SECTION-CUT	7	Continuous	0.50	Yes	Outline of material cut by the section plane - heaviest geometry
T05	ASG-HATCH	252	Continuous	0.09	Yes	Pattern hatches in sections (plots grey)
T06	ASG-HATCH-FILL	251	Continuous	0.09	Yes	Solid fill of thin aluminium / steel walls (plots dark grey)
T07	ASG-SETOUT	8	DASHDOT2	0.13	Yes	Opening lines, setting-out lines that must print
T08	ASG-REFERENCE	253	Continuous	0.13	Yes	Architectural / structural background, xrefs (plots light grey)
T09	ASG-CONSTRUCTION	6	Continuous	0.00	NO	Temporary helper lines only - never final geometry
A01	ASG-DIMENSION	3	Continuous	0.13	Yes	All dimensions
A02	ASG-TEXT	7	Continuous	0.18	Yes	Labels, view titles
A03	ASG-TEXT-NOTE	7	Continuous	0.18	Yes	General and specification notes
A04	ASG-MULTILEADER	3	Continuous	0.13	Yes	Callouts and material labels
A05	ASG-TABLE	7	Continuous	0.18	Yes	Schedules, registers, note tables
A06	ASG-LEVEL	3	Continuous	0.13	Yes	FFL / SSL / sill / head level markers
A07	ASG-GRID	8	CENTER	0.13	Yes	Structural / architectural grid lines and bubbles
A08	ASG-REVISION	1	Continuous	0.25	Yes	Revision clouds and delta tags
A09	ASG-SYMBOL	3	Continuous	0.25	Yes	Section / detail / elevation marks, north arrow
S01	ASG-BORDER	7	Continuous	0.50	Yes	Border frame in layouts
S02	ASG-TITLEBLOCK	7	Continuous	0.25	Yes	Title block insert and attributes
S03	ASG-VIEWPORT	6	Continuous	0.00	NO	All layout viewports
S04	ASG-NOLOT	6	Continuous	0.00	NO	Internal markups and reminders - never printed

ACI	Function	Plotted with ASG_Monochrome.ctb
7	Primary geometry, metals, text, sheet	Black
8	Secondary geometry, hardware, seals, hidden, centrelines, grids	Black (lighter weights)
4	Glass and mirror	Black
3	Annotation: dimensions, leaders, levels, symbols	Black
1	Revisions	Black

ACI	Function	Plotted with ASG_Monochrome.ctb
6	Non-plot: construction, viewport, no-plot	Not plotted (layer plot flag off)
251	Solid fills (aluminium / steel walls in section)	Grey RGB 91 (about 64 % dark)
252	Hatch patterns, insulation	Grey RGB 132 (about 48 % dark)
253	Reference background / xrefs	Grey RGB 173 (about 32 % dark)

6. Lineweight matrix (ISO 128 series) and linetypes

Lineweight	Class	Use	Layers
0.50 mm	Extra heavy	Section cut outlines, sheet border	ASG-SECTION-CUT, ASG-BORDER
0.35 mm	Heavy	Primary visible outlines, cutting planes	ASG-OUTLINE-PRIMARY, ASG-CUTTING-PLANE
0.25 mm	Medium	Aluminium, steel, cladding, symbols, revisions, title block	ASG-ALUMINIUM-PROFILE, ASG-STAINLESS-STEEL, ASG-MILD-STEEL, ASG-CLADDING, ASG-REVISION, ASG-SYMBOL, ASG-TITLEBLOCK
0.18 mm	Light	Secondary lines, glass, hardware, fixings, hidden lines, text, tables	ASG-OUTLINE-SECONDARY, ASG-GLASS, ASG-HARDWARE, ASG-FIXING, ASG-HIDDEN, ASG-TEXT, ASG-TEXT-NOTE, ASG-TABLE
0.13 mm	Fine	Dimensions, leaders, centrelines, grids, seals, insulation, set-out, reference	ASG-GASKET, ASG-SEALANT, ASG-INSULATION, ASG-CENTERLINE, ASG-SETOUT, ASG-REFERENCE, ASG-DIMENSION, ASG-MULTILEADER, ASG-LEVEL, ASG-GRID
0.09 mm	Hatch	Hatch patterns and fills only (also plotted grey)	ASG-HATCH, ASG-HATCH-FILL

Each class is about 1.4 x the next (ISO 128), so the hierarchy survives PDF and paper. 0.09 is used only for hatching, which also plots grey. LTSCALE 1 + PSLTSCALE 1 + MSLTSCALE 1: dash lengths are paper mm in every viewport and in model space.

Linetype	Pattern
HIDDEN	Hidden __ __ __
HIDDEN2	Hidden (.5x) _ _ _
CENTER	Center ____ _ ____
CENTER2	Center (.5x) ____ _ ____
DASHED	Dashed __ __
DASHED2	Dashed (.5x) _ _
DASHDOT	Dash dot __ . __
DASHDOT2	Dash dot (.5x) _ . _
PHANTOM	Phantom ____ _ ____
PHANTOM2	Phantom (.5x) ____ _ ____
DOT	Dot . . .
Continuous	Solid

7. Text system - Arial (TrueType)

Style	Font	Annotative	Use
ASG-TEXT-ANNO	arial.ttf	Yes	Model-space general text and dimension text
ASG-TEXT-NOTE	arial.ttf	Yes	Model-space notes, material labels, leaders
ASG-TEXT-SMALL	arial.ttf	Yes	Small detail notes, profile codes
ASG-TEXT-TITLE	arialbd.ttf	No (paper space)	Detail / view titles (paper space)
ASG-TEXT-HEADING	arialbd.ttf	No (paper space)	Drawing titles, sheet headings (paper space)
ASG-TEXT-SHEET	arial.ttf	No (paper space)	Title block fields, tables, sheet text (paper space)

Use	Plotted height	Style
General dimensions	2.5 mm	ASG-TEXT-ANNO (via dimension style)
Technical notes	2.5 mm	ASG-TEXT-NOTE
Material labels	2.5 mm	ASG-TEXT-NOTE (via ASG-ML-ANNO)
Small detail notes	1.8 mm	ASG-TEXT-SMALL
Detail titles	3.5 mm	ASG-TEXT-TITLE
Drawing titles	5.0 mm	ASG-TEXT-HEADING
Sheet headings	7.0 mm	ASG-TEXT-HEADING (A1); 5.0 on A3 / A4

All heights are from the ISO 3098 series; minimum 1.8 mm on any sheet. Arial is a Windows core font and embeds in PDF; if missing AutoCAD uses FONTALT (set arial.ttf).

8. Dimension system

Style	Annotative	Use
ASG-DIM-ANNO	Yes	Default linear / aligned dimensions in model space, any viewport scale
ASG-DIM-FIXED	No	Non-annotative: dimensions placed in paper space over viewports (true size), or 1:1 sheets
ASG-DIM-DETAIL	Yes	Fabrication details 0.0; symmetrical tolerance switched on per dimension (height 0.7)
ASG-DIM-ANGULAR	Yes	Angular dimensions, decimal degrees 0.00
ASG-DIM-RADIUS	Yes	Radius dimensions (R prefix automatic), no centre mark
ASG-DIM-DIAMETER	Yes	Diameter dimensions (diameter sign automatic), centre mark 2.5
Parameter	Value	
Text	ASG-TEXT-ANNO 2.5, above the dimension line (DIMTAD 1), aligned with it, gap 1.0	
Arrowheads	Closed filled 2.5. Centre mark 2.5 (DIAMETER), none (RADIUS)	
Extension lines	Offset 1.5 from origin, extend 1.25 beyond dimension line	
Spacing	Baseline / continue spacing 7.0	
Fit	DIMATFIT 3 best fit; DIMTOFL 1 line drawn between extension lines; DIMTMOVE 1 leader added when text is moved (RADIUS / DIAMETER: dimension line moves with text)	
Linear units	Decimal mm, '.' separator, precision 0 (DETAIL 0.0), rounding 0, trailing zeros suppressed (DETAIL keeps them: 50.0)	
Angular units	Decimal degrees, 0.00 on ASG-DIM-ANGULAR (trailing zeros shown)	
Tolerance	Off. On ASG-DIM-DETAIL switch on per dimension: symmetrical, precision 0.0, height 0.7	
Alternate units	Off	
Measurement scale	DIMLFAC 1.0 - always true geometry	
Scale	Annotative styles take the annotation scale; ASG-DIM-FIXED DIMSCALE 1 for paper-space dims	
Header	Drawing's current dimension variables equal ASG-DIM-ANNO - no '<style overrides>' on new drawings	

Rule: dimensions always report measured geometry (DIMLFAC 1, DIMASSOC 2). Overriding dimension text is prohibited - fix the geometry. 'Standard' carries ASG-DIM-FIXED values so it is harmless if picked. No further fixed-scale styles are needed.

9. Multileader system

Style	Annotative	Arrow	Text	Frame	Use
ASG-ML-ANNO	Yes	closed filled 2.5	ASG-TEXT-NOTE 2.5	No	Default model-space callouts and material labels
ASG-ML-NOTE	No	closed filled 2.5	ASG-TEXT-SHEET 2.5	No	Paper-space notes pointing into viewports
ASG-ML-DETAIL	Yes	_DOT 1.5	ASG-TEXT-SMALL 1.8	No	Dense 1:1 / 1:2 fabrication details
ASG-ML-REFERENCE	Yes	_OPEN30 2.5	ASG-TEXT-NOTE 2.5	Yes	References to other drawings / details (framed text)

All: straight leader, maximum 2 points, landing gap 1.0, dogleg 4.0, text attached at the middle of the top line, ByLayer colour. Background mask is a property of each note (Properties > Text > Background mask, offset 1.2, use background colour) - use it where a note crosses hatch.

10. Annotative scale governance

Controlled list: 1:1, 1:2, 1:5, 1:10, 1:20, 1:25, 1:50, 1:100. Other scales only with CAD custodian approval (e.g. 1:200 site plans).

Object	Rule
Model-space text	Annotative ASG-TEXT-ANNO / -NOTE / -SMALL - type the paper height (2.5 / 1.8)
Paper-space text	Non-annotative ASG-TEXT-TITLE / -HEADING / -SHEET at true plotted height
Dimensions	ASG-DIM-ANNO family in model space; ASG-DIM-FIXED for dimensions placed in paper space
Multileaders	ASG-ML-ANNO / -DETAIL / -REFERENCE in model space; ASG-ML-NOTE in paper space
Symbols	Section / detail / level / grid bubbles: annotative blocks
Real-size blocks	Never annotative (hardware, profiles, fixings)
Hatches	Material hatches non-annotative (scale rule in section 13); symbolic fills may be annotative

Workflow: 1) Set the viewport scale from the list, then lock the viewport. 2) Work inside the viewport - CANNOSCALE follows it. 3) Add annotation with ASG-*-ANNO styles. 4) Same note needed at another scale: select it > right-click > Annotative Object Scale > Add (OBJECTSCALE). 5) ANNOAUTOSCALE 0 (profile) and ANNOALLVISIBLE 0 (drawing) prevent unwanted scales and visual duplicates; the status-bar 'Annotation Visibility' button shows all scales temporarily. 6) Drawings imported with extra scales: -SCALELISTEDIT > Reset, then AUDIT.

11. Tables and schedules

Style	Title	Header	Data	Use
ASG-TABLE-STANDARD	3.5 MC	2.5 MC	2.5 ML	Standard technical tables
ASG-TABLE-MATERIAL	3.5 MC	2.0 MC	2.0 ML	Material schedules
ASG-TABLE-GLASS	3.5 MC	2.0 MC	2.0 ML	Glass schedules
ASG-TABLE-HARDWARE	3.5 MC	2.0 MC	2.0 ML	Hardware schedules
ASG-TABLE-REVISION	suppressed	2.0 MC	2.0 ML	Revision registers
ASG-TABLE-REGISTER	3.5 MC	2.5 MC	2.5 ML	Drawing registers
ASG-TABLE-NOTES	3.5 ML	suppressed	2.5 TL	General notes

Common: ASG-TEXT-SHEET, cell margins horizontal 1.5 / vertical 1.0, flow top-down, inside grid 0.18, outline 0.35, header fill ACI 254 (light grey), no data fill, colour ByBlock. Alignment codes: MC middle-centre, ML middle-left, TL top-left. Numbers centred, descriptions left. Tables go in paper space on ASG-TABLE. Created by ASG-SETUP.

12. Layouts, page setups, title block and numbering

Layout / named page setup	Paper (canonical media name)	Orient.	Title block	Default viewport
A4-LANDSCAPE / ASG-A4-LANDSCAPE	ISO_full_bleed_A4_(297.00_x_210.00_MM)	L	ASG-TB-A4	1:10, locked
A4-PORTRAIT / ASG-A4-PORTRAIT	ISO_full_bleed_A4_(210.00_x_297.00_MM)	P	ASG-TB-A4	1:10, locked
A3-LANDSCAPE / ASG-A3-LANDSCAPE	ISO_full_bleed_A3_(420.00_x_297.00_MM)	L	ASG-TB-A3	1:20, locked
A3-PORTRAIT / ASG-A3-PORTRAIT	ISO_full_bleed_A3_(297.00_x_420.00_MM)	P	ASG-TB-A3	1:20, locked
A1-LANDSCAPE / ASG-A1-LANDSCAPE	ISO_full_bleed_A1_(841.00_x_594.00_MM)	L	ASG-TB-A1	1:50, locked

All page setups: DWG To PDF.pc3, plot area Layout, scale 1:1 (mm), offset 0,0, ASG_Monochrome.ctb, plot object lineweights ON, lineweights not scaled, plot transparency OFF, paper space last, viewport borders not plotted. Border: 20 mm filing margin left, 10 mm other sides. Viewports on ASG-VIEWPORT, scale from the controlled list, Display Locked = Yes; unlock only to change the scale, then lock again.

Title blocks: separate A4 (180 x 68.2 mm), A3 (180 x 81.9 mm) and A1 (250 x 109.2 mm) designs inserted at scale 1 at the border corner - never stretched. Not explodable, uniform scale, mm units, attributes position-locked (edit values with EATTEDIT). Hierarchy bottom-up: drawing number / revision / sheet; scale / date / status; signatures; drawing title; client / consultant; project / location; company heading (text - no logo invented). Attributes: PROJECT, LOCATION, CLIENT, CONSULTANT, DWG_TITLE, DWG_TITLE_2, DWG_NO, REV, DATE, DRAWN_BY, CHECKED_BY, APPROVED_BY, SCALE, SHEET_NO, DWG_STATUS. Drawing status values: PRELIMINARY, FOR APPROVAL, APPROVED, FOR CONSTRUCTION, FOR FABRICATION, AS-BUILT, SUPERSEDED.

Drawing numbering	
Format	AS-[YYYY]-[NNN]-[DIS]-[TYP]-[SSS] + revision R0, R1, R2 ... in the REV field
Example	AS-2026-014-ALU-SD-001, REV R0 -> file AS-2026-014-ALU-SD-001-R0_MarinaTower.dwg
YYYY-NNN	Al Siraj project code (year + project serial)
DIS	ALU aluminium GLZ glazing MS mild steel SS stainless steel MISC other metal works
TYP	GA general arrangement EL elevation SD shop drawing SEC section DT detail
SSS	Sheet serial 001, 002 ...

13. Hatch, block and attribute standards

Material	Pattern	Angle prop.	Model-space scale	Layer	Note
Glass (section)	ANSI31	0	0.5 x drawing scale (1.6 mm on paper)	ASG-HATCH	Thin monolithic glass may be left unhatched
Aluminium (section)	SOLID	-	-	ASG-HATCH-FILL	Plots dark grey
Mild steel (section)	ANSI32	0	1.0 x drawing scale	ASG-HATCH	Lines read 45 deg
Stainless steel (section)	ANSI31	90	0.75 x drawing scale	ASG-HATCH	Lines read 135 deg
Thermal break / polyamide	ANSI37	0	0.5 x drawing scale	ASG-INSULATION	Cross-hatch
Insulation board / batt	INSUL	0	insulation thickness / 9.525 (real size)	ASG-INSULATION	Pattern band fits the thickness
Concrete	AR-CONC	0	real size: 1.0 up to 1:10, 2.0 at 1:20-1:25, 5.0 at 1:50-1:100	ASG-REFERENCE	
Masonry / blockwork (section)	ANSI31	0	1.0 x drawing scale	ASG-REFERENCE	Plots light grey
Blockwork (elevation)	AR-B816	0	real size 0.984 (= 200 x 400 block)	ASG-REFERENCE	
Sealant bead	SOLID	-	-	ASG-SEALANT	Plots black
Generic / unspecified section	ANSI31	0	1.0 x drawing scale	ASG-HATCH	Use only until material is confirmed

Pattern modules (acadiso.pat, scale 1): ANSI31 / ANSI37 lines 3.175 mm apart, ANSI32 pairs 9.525 mm, INSUL band 9.525 mm, AR-* patterns are real size. 'x drawing scale' patterns keep a constant spacing on paper. All pattern names verified to exist in acadiso.pat.

Blocks: name ASG-<CATEGORY>-<ITEM>-<VARIANT>, geometry on layer 0 ByLayer / ByBlock, base point at a datum (fixing point, bottom-left or centreline), block units mm, uniform scale, real-size items non-annotative, symbols annotative. Manufacturer data only from verified supplier sources.

Attribute tag	Meaning
ITEM_CODE	Item / product code
ITEM_NAME	Item description
SYSTEM_NAME	System / series
MANUFACTURER	Manufacturer / supplier (verified data only)
MATERIAL	Base material
FINISH	Finish
COLOUR	Colour / RAL
GLASS_SPEC	Glass specification
REVISION	Block revision

14. Plotting / PDF standard

Item	Standard
Device	DWG To PDF.pc3 (ships with AutoCAD) - no custom PC3 dependency
Paper	ISO full bleed A4 / A3 / A1 media = exact sheet size; the border provides the margins
Area / scale / offset	Layout, 1:1 (mm), offset 0,0 ('Center the plot' is not available for Layout)
Plot style	ASG_Monochrome.ctb - black; ACI 250-254 keep their grey. Fallback: monochrome.ctb
Lineweights	Plot object lineweights ON, scale lineweights OFF, plot paper space last
Transparency / shading	Plot transparency OFF; shaded viewports 'As displayed', presentation quality
PDF options	Vector 1200 dpi, raster 300 dpi, TrueType as text, layer information OFF for client issue
Pre-issue	AUDIT, PURGE, plot preview, open the PDF and check scale, dimensions and no-plot items
Compatibility	AutoCAD 2018 DWG (opens in AutoCAD 2018-2026); DXF 2018 for third parties; no proxies
Xrefs / PDF import	Relative paths, overlay, layer ASG-REFERENCE; PDFIMPORT geometry moved to ASG layers

15. Resource separation

Resource	Where it lives	Deploy
Layers, styles, scales, layouts, title blocks, header variables	Inside the DWT	Copy DWT to shared Templates folder
Table styles, named page setups, CANNOSCALE / MSLTSCALE / HPLAYER / DIMLAYER ...	Inside the DWT after ASG-SETUP	Travel with the DWT
ASG_Monochrome.ctb	Plot Styles folder (external)	Copy to each PC or a shared support path
DWG To PDF.pc3	Plotters folder (ships with AutoCAD)	None
arial.ttf / arialbd.ttf	Windows Fonts (external)	Present on Windows; FONTALT = arial.ttf as fallback
Linetypes	Embedded in the DWT (acadiso values)	None
Hatch patterns	acadiso.pat (ships with AutoCAD)	None
OSMODE, grips, autosave, pickbox ...	User profile / registry	Optional ASG-PROFILE per user
QNEW default template	Options > Files (profile)	Set per PC to ASG_Master_Template.dwt

16. Portability checklist (each PC)

- [] AutoCAD 2018 or later installed; DWG To PDF.pc3 present (default).
- [] ASG_Monochrome.ctb copied to the Plot Styles folder - ASG-QA confirms it is found.
- [] ASG_Master_Template.dwt on the shared Templates path; QNEW points to it.
- [] Arial installed (Windows default); FONTALT = arial.ttf.
- [] Open a new drawing from the DWT and run ASG-QA: RESULT must be 0 FAIL.
- [] Optional: user runs ASG-PROFILE once (snaps, grips, autosave).

17. Finishing and verifying in AutoCAD

1. Copy ASG_Monochrome.ctb to the Plot Styles folder; open it in the Plot Style Table Editor: Lineweight column must read 'Use object lineweight', Color = Black (250-254 'Use object color').
2. OPEN ASG_Master_Template.dxf, AUDIT Y, APPLOAD ASG_Setup.lsp, run ASG-SETUP.
3. Read RESULT line / ASG_QA_Log.txt: every line PASS.

4. SAVEAS AutoCAD 2018 DWG, then SAVEAS DWT (Measurement Metric).
5. Open QA_TEST_DRAWING_temporary.dxf, plot A3-LANDSCAPE to PDF: check dims, no 'NOPLOT' text / diagonal, no viewport frames, 6 lineweights distinguishable, text legible.
6. In a new drawing from the DWT: annotative text / dim / leader at 1:20 then add 1:5 via OBJECTSCALE - confirm 2.5 mm on paper in both viewports. Do not save the test.

Appendix - layout previews (rendered from the saved master DXF)

A4-LANDSCAPE

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.			
AL SIRAJ GROUP ALUMINIUM, GLASS & FACADE DIVISION UNITED ARAB EMIRATES			
PROJECT		LOCATION	
PROJECT NAME		LOCATION, UAE	
CLIENT		CONSULTANT	
CLIENT		CONSULTANT	
DRAWING TITLE			
DRAWING TITLE			
DRAWN BY		CHECKED BY	APPROVED BY
-		-	-
SCALE		DATE	DRAWING STATUS
AS SHOWN		DD.MM.YYYY	PRELIMINARY
DRAWING No.		REV	SHEET No.
AS-YYYY-NNN-DIS-TYP-001		R0	01 OF 01

A4-PORTRAIT

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.			
AL SIRAJ GROUP ALUMINIUM, GLASS & FACADE DIVISION UNITED ARAB EMIRATES			
PROJECT		LOCATION	
PROJECT NAME		LOCATION, UAE	
CLIENT		CONSULTANT	
CLIENT		CONSULTANT	
DRAWING TITLE			
DRAWING TITLE			
DRAWN BY		CHECKED BY	
-		-	
SCALE		DATE	
AS SHOWN		DD.MM.YYYY	
DRAWING No.		REV	
AS-YYYY-NNN-DIS-TYP-001		R0	
		SHEET No.	
		01 OF 01	

A3-LANDSCAPE

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.			
AL SIRAJ GROUP ALUMINIUM, GLASS & FACADE DIVISION UNITED ARAB EMIRATES			
PROJECT		LOCATION	
PROJECT NAME		LOCATION, UAE	
CLIENT		CONSULTANT	
CLIENT		CONSULTANT	
DRAWING TITLE			
DRAWING TITLE			
DRAWN BY		CHECKED BY	APPROVED BY
-		-	-
SCALE		DATE	DRAWING STATUS
AS SHOWN		DD.MM.YYYY	PRELIMINARY
DRAWING No.		REV	SHEET No.
AS-YYYY-NNN-DIS-TYP-001		R0	01 OF 01

A3-PORTRAIT

A1-LANDSCAPE

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.		
AL SIRAJ GROUP ALUMINIUM, GLASS & FACADE DIVISION UNITED ARAB EMIRATES		
PROJECT	LOCATION	
PROJECT NAME	LOCATION, UAE	
CLIENT	CONSULTANT	
DRAWING TITLE		
DRAWING TITLE		
DRAWN BY	CHECKED BY	APPROVED BY
DATE	DATE	DRAWING STATUS
AS SHOWN	DD.MM.YYYY	PRELIMINARY
DRAWING NO.	REV	SHEET NO.
AS-YYYY-NNN-DIS-TYP-001	R0	01 OF 01

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.		
AL SIRAJ GROUP ALUMINIUM, GLASS & FACADE DIVISION UNITED ARAB EMIRATES		
PROJECT	LOCATION	
PROJECT NAME	LOCATION, UAE	
CLIENT	CONSULTANT	
CLIENT	CONSULTANT	
DRAWING TITLE		
DRAWING TITLE		
DRAWN BY	CHECKED BY	APPROVED BY
+	+	+
SCALE	DATE	DRAWING STATUS
AS SHOWN	DD.MM.YYYY	PRELIMINARY
DRAWING NO.	REV	SHEET NO.
AS-YYYY-NNN-DIS-TYP-001	R0	01 OF 01